Course notes

Set a sequence and calculate

A real sequence is a function defined on $\mathbb{N}$, or on integers from a row n_0. We note u_n his term of rank n. An explicit formula gives u_n direct; a recurrence relationship requires also an initial term. $u_0 = 2$ and $u_{n+1} = 3u_n - 1$ give $u_1 = 5$, $u_2 = 14$.

Reference sequences

If $u_{n+1} = u_n + r$, then $u_n = u_p + (n-p)r$. If $u_{n+1} = qu_n$, then $u_n = u_p q^{n-p}$ for $n \geq p$.

$$\sum_{k=p}^{n} u_k = (n-p+1)\frac{u_p + u_n}{2} \quad \text{(arithmetic)}, \qquad \sum_{k=p}^{n} u_k = u_p \frac{1-q^{n-p+1}}{1-q} \quad (q \neq 1).$$

When $q = 1$, the second sum is $(n-p+1)u_p$.

Induction reasoning

To prove $P(n)$ for the integer $n \geq n_0$: check $P(n_0)$; fix an arbitrary integer $n \geq n_0$ and assuming $P(n)$ true; deduce from it $P(n+1)$; the conclusion by the principle of recurrence. not the conclusion: it can only be used at the fixed rank.

Completely written example. Let $u_0 = 1$, $u_{n+1} = 2u_n + 3$. For $n \in \mathbb{N}$,note $P(n)$: « $u_n = 4 \times 2^n - 3$ ».

Initialization. $4 \times 2^0 - 3 = 1 = u_0$; $P(0)$ is true.

Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True. By the definition of the future, then by this hypothesis,

$$\begin{aligned} u_{n+1} &= 2u_n + 3 \\ &= 2(4 \times 2^n - 3) + 3 \\ &= 8 \times 2^n - 3 = 4 \times 2^{n+1} - 3. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. By the principle of recurrence, $u_n = 4 \times 2^n - 3$ for all $n \in \mathbb{N}$.

Demonstration: sum of integers. Note: $P(n)$: « $\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$ For $n = 0$, both members are 0: $P(0)$ It's true. $n \geq 0$ and suppose $P(n)$ True, then.

$$\begin{aligned} \sum_{k=0}^{n+1} k &= \sum_{k=0}^{n} k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

We've established $P(n+1)$. By induction, the formula is true for everything $n \in \mathbb{N}$.

Overall behaviour

The rest is increasing if $u_{n+1} \geq u_n$ for any rank, decreasing if inequality is reversed. increased if there is a real M such that $u_n \leq M$ for every n ; minor if it exists m such that $u_n \geq m$; limitate if it is both reduced and increased. To study monotonicity, one can study the sign of $u_{n+1} - u_n$. If all the terms are strictly positive, we can compare $\frac{u_{n+1}}{u_n}$ 1. The study of a function f such that $u_n = f(n)$ is another possibility.

Arithmetico-geometric sequence

For $u_{n+1} = au_n + b$, $a \neq 1$, solve $\ell = a\ell + b$, then pose $v_n = u_n - \ell$. We get $v_{n+1} = av_n$, so

$$u_n = \ell + (u_0 - \ell)a^n, \qquad \ell = \frac{b}{1-a}.$$

If $a = 1$, the sequence is arithmetic: do not divide by 1 $- a$.

Example. If $u_0 = 10$ and $u_{n+1} = 0{,}8u_n + 6$, then $\ell = 30$ and $u_n = 30 - 20(0{,}8)^n$. We have $10 \leq u_n < 30$ and $u_{n+1} - u_n = 4(0{,}8)^n > 0$.

Trap. Checking a property on a hundred terms doesn't show it. A increasing sequence isn't necessarily positive; a bounded sequence is not necessarily monotonous.